

Improving the safety of navigation and the sustainability of shipping through the introduction of innovative autonomous shipping technologies in the Asia-Pacific region

Study Report on regional cooperation on autonomous shipping in Asia and the Pacific

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I. INTRODUCTION

The project on "Improving the Safety of Navigation and the Sustainability of Shipping through the Introduction of Innovative Autonomous Shipping Technologies in the Asia-Pacific Region" is implemented in accordance with the ESCAP programme of work approved in 2022 for subprogramme 3 - Transport. This initiative aims to achieve the overarching objective of sustainable transport connectivity, logistics, and mobility in the Asia-Pacific region, as well as the generation and transfer of knowledge.

The project conforms to the means of implementation and partnership identified in the Regional Roadmap. It specifically contributes to enhancing technology and policy coherence by facilitating the communication of best practices and capacity-building across member countries (para 16 (a)) and promoting integrated policies based on the systems approach and other methodologies (para 20 (c)).

The project directly aligns with the objectives described in paragraph 32(a) of the Regional Roadmap, which focuses on developing and integrating maritime connectivity, implementing regional transport facilitation frameworks, and establishing technical standards for operationalizing transport connectivity. Additionally, this project will contribute to implementing several thematic components of the Regional Action Programme for Sustainable Transport Development in Asia and the Pacific (2022-2026), including "Maritime and Interregional Transport Connectivity," "Digitalization of Transport," and "Low Carbon Mobility and Logistics."[1]

The project is expected to enhance countries' capabilities of implementing innovative autonomous shipping technologies through the exchange of experiences and best practices and to strengthen regional cooperation. It aims to identify challenges and opportunities created by autonomous shipping technologies, highlighting legal and technical obstacles as well as the broader benefits of these technologies. Based on this analysis, national workshops will be held in selected countries to develop national implementation plans. At the regional level, expert meetings and consultations will be organized to develop a regional approach to the adoption of innovative autonomous transport technologies in Asia and the Pacific and to identify opportunities for joint pilot projects among ESCAP member States.

[1] Regional Action Programme for Sustainable Transport Development in Asia and the Pacific (2022–2026), UN ESCAP, 2022, https://repository.unescap.org/handle/20.500.12870/4669?locale-attribute=es

One of the key outputs of the project is the development of national plans for the implementation of autonomous shipping technologies in the target countries and consideration of initiatives for regional collaboration. A study on international developments in autonomous shipping has been prepared, along with national-level study reports for each of the project's target countries (India, Indonesia, Malaysia, Thailand, and Viet Nam). These studies offer practical recommendations for the introduction of autonomous shipping based on an analysis of the situation in the target countries and the entire Asia-Pacific region.

The purpose of this study report is to summarize reports and expert opinions from national workshops held in 2023 and the International Conference on Autonomous Shipping in February 2024. It proposes policy recommendations for consideration and adoption at the Regional High-Level Dialogue on Sustainable Maritime Connectivity, organized pursuant to ESCAP Commission Resolution 76/1. The report also highlights promising joint projects and their impact on ensuring the region's leading position in the safety and efficiency of shipping and the development of advanced marine technologies, as well as technical solutions for the introduction and application of autonomous shipping.

II. TARGET COUNTRIES' EXPERIENCES

A. READINESS FOR COOPERATION ON AUTONOMOUS SHIPPING OF TARGET ASIA-PACIFIC COUNTRIES

The Asia-Pacific region is emerging as a significant hub for the development and deployment of autonomous shipping technologies. These efforts are driven by a combination of academic, industrial, and governmental initiatives aimed at modernizing maritime infrastructure, enhancing operational efficiency, and ensuring compliance with international standards. The region's strategic geographical position and extensive maritime activities provide a fertile ground for the growth of autonomous shipping.

Countries like India, Indonesia, Malaysia, Thailand, and Viet Nam have undertaken various initiatives and projects to integrate autonomous shipping technologies into their maritime operations. At the same time, currently these projects are limited by small size and scale model vehicles, including Navy (military) purpose vehicles.

In the target countries, the major role in the development and adoption of autonomous shipping is played by government agencies and academic and research institutions. The government agencies are leading efforts to develop regulatory frameworks and support technological infrastructure, while universities are at the forefront of research and training for autonomous shipping. It is important to highlight that these efforts are aligned with international organizations (mainly the International Maritime Organization (IMO)) and sometimes involve foreign research and technology partners.

The readiness for cooperation on autonomous shipping in the Asia-Pacific region is underscored by several common features, presented below:

- **Regulatory Alignment:** Countries are actively aligning their regulatory frameworks to conform to international standards, such as the IMO's MASS (Maritime Autonomous Surface Ship) Code, to ensure the safe and responsible deployment of autonomous shipping technologies.
- **Technological Infrastructure:** Significant investments are being made in technological infrastructure, including smart port and e-navigation development.
- **Collaborative Research and Development:** There is a strong emphasis on collaboration between academic institutions, research organizations, and industry players to drive innovation in autonomous shipping.
- **Skill Development:** Training programs and initiatives are being prioritized to prepare seafarers and maritime professionals for the integration of autonomous systems into their operations.

A common feature is the weak **engagement of the private sector** (shipping companies, shipbuilders, and technology firms), which has to be the key player in implementing and testing autonomous shipping technologies. One of the main challenges associated with the private sector is the high upfront investment required to develop, integrate, and deploy autonomous technologies.

Focusing on these common features, the target countries together with the support of other countries of the Asia-Pacific region and the secretariat of the Economic and Social Commission for Asia and the Pacific (ESCAP) can develop joint efforts to speed up the implementation of the new technology while mitigating risks and decreasing required investments, especially by the private sector of the target countries.

B. INDIA: NATIONAL EXPERIENCES

Autonomous shipping technologies and infrastructure developments in India

India has made significant progress in developing autonomous shipping technologies and has established a regulatory framework to support their implementation[2]. The Indian government has recognized the potential of autonomous shipping to enhance sustainability and efficiency in the maritime sector.

One of the most valuable governmental initiatives in the maritime industry is Sagarmala, India's flagship program for maritime progress. This program has outlined over 200 port connectivity projects to improve access to ports via road, rail, coastal, and inland routes. Sagarmala aims to help India achieve its target of sourcing 40% of national energy from renewable sources by 2030.

Two major roadmaps under Sagarmala have been adopted by the Government of India: the Maritime India Vision (MIV) 2030[3], which aims to develop best-in-class port infrastructure, and the Maritime Amritkaal Vision 2047 (previously referred to as the Blue Economy Vision 2047)[4], which specifically identifies the development of autonomous shipping as a key priority. The theme includes an Implementation Plan for Infrastructure, Institutional Capacity Building, and Technology Initiatives. This includes the establishment of an Advanced Nautical Studies Center for new technologies in navigation, such as autonomous underwater instrumentation, along with technology adaptation education and training using augmented and virtual reality (AR and VR) for seafarers, and regulatory changes for new technologies.

Under this roadmap, the Center for Excellence in Maritime and Shipbuilding (CEMS) was established to design courses based on industry needs and create mentorship programs. CEMS aims to facilitate simulator support in training, which could include programs on autonomous shipping in India.

In the framework of the Sagarmala Programme on March 22, 2023, the Government of India has created the National Centre of Excellence for Green Port

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[2] Development of the autonomous shipping technologies, 2024, United Nation ESCAP, https://www.unescap.org/sites/default/d8files/event-documents/India_national-study-report-on-autonomous-shipping.pdf

[3] The Maritime India Vision (MIV) 2030, 2021, Ministry of Ports, Shipping and Waterways Government of India, https://sagarmala.gov.in/sites/default/files/MIV%202030%20Report.pdf

[4] Maritime Amrit Kaal Vision 2047, 2023, Ministry of Ports, Shipping and Waterways Government of India, https://shipmin.gov.in/sites/default/files/Maritime%20Amrit%20Kaal%20Vision%202047%20%28MAKV%202047%29_compressed_0.pdf

& Shipping (NCoEGPS) with the aim to develop an alternate technology adoption roadmap for green shipping in India. The Centre could be used to offer innovation labs and incubation facilities for startups in autonomous shipping.

The Center for Advanced Nautical Studies at the Indian Maritime University also conducts research in autonomous shipping. The center is crucial in creating facilities for technology adaptation, education, and the creation and training of innovation labs.

The Ministry of Science and Technology of India supports research on autonomous shipping. Through initiatives like the National Mission on Interdisciplinary Cyber-Physical Systems (NMICPS), the Department of Scientific and Industrial Research (DSIR) funds and supports technological innovation hubs such as the Technology Innovation Hub on Autonomous Navigation (TiHAN) Foundation at Indian Institute of Technology Hyderabad, which focuses on autonomous shipping[5].

Legislation on autonomous shipping in India

Current regulations for autonomous shipping in India are guided by the Directorate General of Shipping (DGS), the regulatory body for maritime affairs. The Indian Register of Shipping (IRCLASS) has issued guidelines for the classification of remotely operated and autonomous surface vessels, initially published in December 2021 and revised later in December 2022. These guidelines address various aspects of autonomous vessels, including design, operation, maintenance, data privacy, and cybersecurity, providing standards to ensure safety and regulatory compliance. The regulatory framework in India is poised to accommodate the development of autonomous shipping technologies. The DGS coordinates with the IMO to ensure alignment with international regulations.

Potential routes and projects for the use of autonomous shipping technologies

India has several potential routes and projects in which autonomous shipping technologies may be used to strive towards environmental sustainability and efficiency of shipping services.

Inland Waterways: Autonomous vessels can be used to reduce urban road congestion by facilitating cargo movement along National Waterways, such as the upgraded infrastructure of the National Waterway. MASS can facilitate the modal shift of longitudinal movement of cargo along National Waterways in the

[5] TiHAN-IIT Hyderabad, 2023, the site of the TiHAN-IIT (checked at 20.05.2024), https://tihan.iith.ac.in/tihan-foundation-2/

upgraded infrastructure of National Waterway 1 in the states of Uttar Pradesh, West Bengal, Jharkhand, and Bihar. The cargo movement could energize economic activities, impacting ecosystems along the riverbank and acting as a catalyst to boost inclusive growth, and further supplement and strengthen existing modes of transportation as decided by the Government.

Coastal Shipping: Autonomous ships can be deployed for point-to-point movement between ports, such as between the Jawaharlal Nehru Port and the Mumbai Port, reducing emissions and road congestion.

Sundarbans: The use of autonomous vessels near the fragile environment of the Sundarbans may reduce soil erosion and threats to the crew from hostile fauna.

The adoption of autonomous shipping technologies in the Indian shipping industry will hold significant potential for improving safety, efficiency, and environmental sustainability. By investing in advanced marine technologies and fostering collaboration with global leaders, India can enhance its maritime capabilities and contribute to the broader goals of sustainable and resilient maritime transport.

Challenges and issues in implementing autonomous shipping technologies

India has made significant strides in the field of autonomous shipping, but several challenges must be addressed to ensure its successful implementation: [6]

Collision avoidance: Ensuring that autonomous vessels can accurately avoid collisions is crucial, particularly in the densely trafficked and unpredictable weather conditions of the Southeast Asian region.

Narrow channels and traffic schemes: Inland waterways often feature narrow channels. It is essential to develop algorithms that consider these narrow passages, Traffic Separation Scheme (TSS), and fairways.

Obstacle detection: Developing algorithms capable of accurately detecting and navigating around fishing vessels, small boats, and other common obstacles in Indian waterways is necessary.

Seafarer employment and training: The impact on seafarer employment must be addressed, along with the need for training and upskilling the maritime workforce to operate and maintain autonomous systems.

[6] Bornali Rahman, Analyzing appropriate autonomous vessel for South-East Asian route: from the view of seafarers, Journal of Shipping and Trade volume 7, Article number: 21 (2022), https://jshippingandtrade.springeropen.com/articles/10.1186/s41072-022-00122-9

Development and testing facilities: The implementation of autonomous ships is a developing field that requires comprehensive facilities for development and testing, including machinery, manufacturing, algorithm development, sensors, simulation, and training.

Customer acceptance: Gaining customer acceptance and overcoming resistance to the use of unmanned vessels is essential for the widespread adoption of autonomous shipping technologies.

Policy recommendations for India

Establish a Regulatory Framework: Develop and implement a comprehensive regulatory framework ensuring safety, security, and adherence to international norms, regularly updated to incorporate new technologies and best practices.

Training and Skilling: Provide training programs for seafarers to adapt to new technologies and new roles in autonomous shipping.

Phased Implementation: Implement autonomous shipping technologies in phases, starting with routes currently unserved by maritime transport and routes where establishing autonomous shipping would not involve significant job losses.

Public Engagement and Awareness: Engage with the public and stakeholders through regular updates and educational initiatives to ensure understanding and address challenges.

Collaboration and Partnerships: Foster partnerships between government agencies, industry stakeholders, and research institutions to leverage expertise and resources.

Investment in Infrastructure: Invest in the development of infrastructure supporting autonomous shipping, including advanced sensors, communication systems, and data analytics capabilities.

Insurance and Liability: Develop insurance policies covering risks associated with autonomous shipping, including damage to vessels, cargo loss, and other liabilities.

Monitoring and Evaluation: Establish a monitoring and evaluation framework to track progress and impact, identifying areas of improvement.

International Cooperation: Engage with international organizations like the IMO to share best practices, collaborate on research and development, and ensure alignment with global standards.

Research and Development: Encourage research and development in autonomous shipping technologies through funding and partnerships.

Overcoming Barriers: Identify and address barriers to the adoption of autonomous shipping, including job loss concerns, technology acquisition, and financing.

Public-Private Partnerships: Foster public-private partnerships to leverage expertise and resources.

Capacity Building: Build capacity within the Indian maritime sector to design, develop, and operate autonomous vessels. This includes training programs for engineers, technicians, and other professionals.

Cybersecurity: Ensure that the infrastructure and systems supporting autonomous shipping are secure from cyber threats. This includes implementing robust cybersecurity measures and conducting regular security audits.

Environmental Impact: Assess and mitigate the environmental impact of autonomous shipping, including reduced emissions and noise pollution. This can be achieved through the use of green technologies and sustainable practices.

By addressing these challenges and implementing the recommended policies, India can position itself as a leader in the development and deployment of autonomous shipping technologies, enhancing its maritime capabilities and contributing to global sustainable transport goals.

Promising ideas of autonomous shipping projects in India

Implementation of joint trial project with the Cochin Shipyard

The Cochin Shipyard is the first Indian company that started building ships for autonomous shipping but still was not involved in the implementation or any trials of the new technologies. The leading technology companies that are involved in the major projects of autonomous shipping trials such as the Autonomous and Remote Navigation Trial Project (ARNTP), MEGURI 2040, or other extensive trials projects have substantial experience in developing and testing autonomous maritime technologies. By collaborating with Cochin Shipyard, these companies can share their expertise, helping to accelerate the development of autonomous shipping systems in India. The integration of autonomous navigation systems can make ship operations more efficient. Autonomous shipping technologies have demonstrated significant improvements in operational efficiency and safety during trials. Implementation of autonomous technologies in India can help reduce labor costs and improve fuel efficiency, leading to significant cost savings. The extensive trials and risk assessments conducted by technology companies can help identify and mitigate potential risks associated with autonomous shipping, ensuring safer maritime operations.

Joint training programs for seafarers and joint research with the IMU

The Indian Maritime University (IMU) is a central institution dedicated to maritime education, training, and research in India. IMU is involved in various research projects focused on advancing autonomous shipping technologies. This includes studies on the integration of AI, machine learning, and sensor technologies in maritime operations. The university's simulation facilities could be used for testing and validating new autonomous shipping technologies, ensuring that they meet the required safety and performance standards before deployment[7]. Cooperation with other universities, serving at the forefront of developing training programs for autonomous shipping, can bring IMU their advanced curricula and training methodologies to enhance the quality of education at IMU. Through educational programs, research initiatives, industry collaborations, and public engagement efforts in India, IMU could be at the forefront of driving innovation and excellence in the Indian maritime sector.

[7] The brief history of maritime training & education in India, Chapter 17, Directorate General of Shipping, Ministry of Ports, Shipping and Waterways Government of India, https://www.dgshipping.gov.in/Content/PageUrl.aspx?page_name=ShipManualChap17

C. INDONESIA: NATIONAL EXPERIENCES

Autonomous shipping technologies in Indonesia

Indonesia, as the world's largest archipelago, is uniquely positioned to benefit from autonomous shipping technologies. The Indonesian government has been proactive in exploring these technologies to enhance the efficiency and safety of its vast maritime sector.

Indonesia's Ministry of Transportation has initiated several projects aimed at modernizing maritime infrastructure, including the development of autonomous shipping capabilities. These initiatives align with the country's "Sea Toll" program, which aims to improve logistics and reduce disparities between the western and eastern regions of Indonesia.

One significant project is the establishment of the Batam Maritime Industrial Center (BMIC), which includes a dedicated research and development hub for autonomous shipping technologies. This center collaborates with leading international technology providers to develop and test autonomous vessels, focusing on local and regional applications.

Indonesia has also embarked on developing smart ports, integrating digital technologies to enhance port operations' efficiency and safety. The Port of Tanjung Priok, Indonesia's busiest port, is undergoing significant upgrades to include autonomous and remotely operated systems. There are two examples of developments of autonomous shipping in Indonesia: Firstly, PT PAL Indonesia (Persero) is developing an unmanned or autonomous submarine in cooperation with a German company, Thyssen-Krupp Marine Systems, to meet Indonesia's underwater defense needs. The Institut Teknologi Sepuluh Nopember (ITS) collaborated with Robo Marine Indonesia (RMI) and the Naval Technology College (STTAL) to launch an autonomous ship innovation. This Autonomous Modular Unmanned Surface Vehicle (USV) is designed to map ocean conditions in Indonesia and can be used for various aspects in the Indonesian maritime sector[8].

This institution launched an innovative unmanned autonomous Intelligent Boat (I-Boat) in 2020. This unmanned vessel utilizes a combination of artificial intelligence and Internet of Things technologies. It can identify potential dangers and prevent collisions with objects at sea. Moreover, it can reduce the risk of ship

[8] Collaborating with Two Well-known Agencies, ITS Launches Autonomous Ship Innovation, Dec 13, 2023, https://www.its.ac.id/news/en/2023/12/13/collaborating-with-two-well-known-agencies-its-launches-autonomous-ship-innovation/

accidents and optimize its propulsion. The I-Boat can be controlled through a user interface application or a mobile phone. The operator can steer the vessel to sail toward the desired coordinates.

Legislation on autonomous shipping in Indonesia

There are no specific laws or regulations in Indonesia addressing the adoption of autonomous shipping. The Ministry of Transport, particularly the Directorate General of Sea Transport (DGST), is responsible for regulating the maritime sector. However, the country lacks a comprehensive regulatory framework for autonomous shipping, and it does not support its widespread adoption.

The Indonesian government is considering potential initiatives to promote the adoption of autonomous shipping technology. These initiatives include collaborations between government entities, industry stakeholders, and academia to drive the development and deployment of autonomous shipping in Indonesia. The Ministry of Education and Research is also involved in promoting research and development in autonomous shipping.

Indonesia's legal framework for autonomous shipping is being developed in consultation with the International Maritime Organization (IMO) to ensure alignment with global regulations. The country aims to create a comprehensive regulatory environment that supports innovation while ensuring maritime safety and environmental protection.

Potential of adoption of MASS in Indonesia

Indonesia's extensive archipelago presents numerous opportunities for deploying autonomous vessels:

- **Inter-Island Cargo Transport**: Autonomous vessels can enhance logistics by connecting remote islands with major ports, reducing transportation costs and improving supply chain efficiency.
- **Tourism and Passenger Ferries:** Autonomous ferries can provide safe and efficient transport for passengers, particularly in regions with high tourism potential, such as Bali and the Raja Ampat Islands.
- **Environmental Monitoring:** Autonomous vessels can be used for environmental monitoring and protection in marine conservation areas, helping to preserve Indonesia’s rich biodiversity.

Indonesia is the world's third-largest country **in training and supplying seafarers**. More than 80 seafarers' academies are spread throughout the main islands. The current curriculum of seafarers' academies in Indonesia does not include any advanced subjects related to autonomous shipping. The human

element is crucial in the operation of MASS. The academies need to adapt their curriculum to start changing the landscape of automation in shipping. According to Ms. Hafida Fahmiasari's study[9], several subjects need to be added to the curriculum: Autonomous Systems, Data Analytics and Artificial Intelligence, Cybersecurity, Human-Machine Interaction, Regulatory and Legal Frameworks, and Ethical and Social Implications.

Barriers to the introduction of autonomous shipping in Indonesia include the lack of regulatory frameworks, lack of awareness of the new autonomous shipping technology, high initial costs, and the need for significant investments in infrastructure and human resources. Additionally, there are public concerns about the potential impact on employment and the need for seafarers to adapt to new roles in the industry.

One of the main challenges associated with a potential economy of scale in MASS operations is the high upfront investment required to develop, integrate, and deploy autonomous technologies. The initial costs of retrofitting existing vessels or constructing new autonomous vessels with advanced systems, sensors, and communication infrastructure can be significant. Additionally, there could be expenses related to the implementation of supporting infrastructure, such as shore-based control centers, data management systems, and training facilities.

The high upfront investment required for MASS operations presents a challenge. Nevertheless, Indonesia has opportunities to overcome this obstacle and benefit from adopting autonomous technologies in the maritime industry. Potential opportunities include, but are not limited to:

• **Technology Localization:** Indonesia can explore opportunities to localize the production and integration of autonomous systems and components. By establishing partnerships with technology providers and encouraging the development of domestic capabilities, the country can reduce reliance on imports and potentially lower the costs associated with implementing MASS technologies.

• **Public-Private Partnerships**: Collaboration between the government, private sector, and research institutions can lead to public-private partnerships focused on financing, researching, and deploying MASS technologies. These partnerships

[9] Hafida Fahmiasari, National study report on autonomous shipping – Indonesia, Materials for the Autonomous Shipping in Asia and the Pacific International Conference, 2024, United Nations ESCAP, https://www.unescap.org/sites/default/d8files/event-documents/Indonesia_national-study-report-on-autonomous-shipping.pdf

can help pool resources, share risks, and leverage expertise to overcome financial barriers and promote the adoption of autonomous systems.

• **Use existing infrastructure:** Indonesia can capitalize on its existing maritime infrastructure and capabilities to facilitate the adoption of MASS. By retrofitting existing vessels with autonomous systems or leveraging established shipbuilding facilities, the country can optimize its resources and minimize the need for extensive infrastructure development.

Policy recommendations for Indonesia

To successfully introduce autonomous shipping in Indonesia, the government should prioritize the development of a comprehensive regulatory framework, invest in infrastructure and human resources, and promote public awareness and education. The government should also collaborate with industry stakeholders and academia to drive research and development in autonomous shipping.

A draft national plan of implementation of autonomous shipping in Indonesia should focus on the following key milestones:

1. Short-term (1-3 years): Education and raising awareness, regulatory framework alignments, trial projects and data collection.

2. Mid-term (4-7 years): National regulations and guidelines, infrastructure upgrades, and capacity building.

3. Long-term (8+ years): An autonomous shipping corridor, continuous regulatory developments, and internal collaborations.

By following this roadmap, Indonesia can effectively integrate autonomous shipping technologies into its maritime industry, enhancing efficiency, safety, and sustainability while addressing the challenges and barriers that come with this transition.

Promising ideas of projects in autonomous shipping in Indonesia

Trial project with the Institute Teknologi Sepuluh Nopember

The Institute Teknologi Sepuluh Nopember is a pioneer in MASS research in Indonesia and has been involved in developing autonomous surface ships for various applications, including search and rescue and hydro-oceanographic mapping. ITS has signed a Memorandum of Understanding (MoU) with PT Pelayaran Nasional Indonesia (PELNI) to support the development of the maritime sector. This collaboration includes the modification of PELNI's passenger ships to accommodate containers and cargo vessels, and the implementation of various ITS innovations such as autonomous life rafts and

remotely operated life buoys[10]. The ITS developments in collaboration with PELNI is a good start for the cooperation of technologies of companies and universities already experienced in autonomous shipping trials in other countries.

PELNI's involvement in various technological and collaborative projects indicates its readiness to embrace autonomous shipping technologies. The company's strategic partnerships, particularly with ITS, and its commitment to innovation and digital transformation, position it as a key player in the potential implementation and trials of autonomous navigation systems in Indonesia.

This project and the ones that are soon to follow are expected to contribute to the growth and development of Indonesia's maritime sector, enhancing its economic competitiveness and sustainability while improving safety and efficiency in maritime operations.

[10] Continue to Downstream, ITS Signs MoU with PELNI, 2022, site of the ITS (checked 20.05.2024), https://www.its.ac.id/news/en/2022/06/22/continue-to-downstream-its-signs-mou-with-pelni/

D. MALAYSIA: NATIONAL EXPERIENCES

Autonomous shipping technologies in Malaysia

Malaysia has been actively involved at the public activities of civil autonomous shipping technologies. The Ministry of transport and Malaysian Marine Department has participated in international conferences and workshops, such as the ESCAP, Asia-Pacific Economy Cooperation forum (APEC) and APEC Port Services Network (APSN) meetings and conferences, to stay updated on the latest advancements and best practices in the sphere of autonomous shipping[11]. Malaysia has experience in utilizing autonomous navigation ships in the defense sphere: the Malaysian Maritime Enforcement Agency (MMEA) has ordered the construction of three Tun Fatimah-class offshore patrol vessels, which are being built by the Malaysian company THHE-Destini with assistance of the Dutch company Damen. These vessels are designed to incorporate advanced navigation and surveillance technologies, including autonomous systems. The Marine department of Malaysia has been working to establish a regulatory framework for autonomous shipping.

Malaysia is aligning its regulatory framework with the International Maritime Organization (IMO)'s Maritime Autonomous Surface Ships (MASS) Code. The MASS Code, expected to be adopted by 2026 and made mandatory by 2032[12], aims to establish an international framework for the regulation of autonomous vessels. The Ministry of Transport of Malaysia is developing national regulations and guidelines for autonomous ships operating in Malaysian waters. These regulations will address safety, security, privacy, and liability issues, ensuring that autonomous shipping technologies are deployed safely and responsibly.

Malaysia is investing in the necessary technological infrastructure to support the development and operation of autonomous ships. This includes advanced navigation systems, collision avoidance systems, electronic chart displays, and automated identification systems.[13]

The Ministry of Transport Malaysia has invested in technological infrastructure to support the development and operation of autonomous ships, including the

[11] Port and MASS: Practices, Challenges and Harmonization, APSN, https://apsn.glueup.cn/event/port-and-mass-practices-challenges-and-harmonization-40442/?pk_campaign=widget-event-list

[12] Maritime Safety Committee - 108th session (MSC 108), 15-24 May 2024, IMO, https://www.imo.org/en/MediaCentre/MeetingSummaries/Pages/MSC-108th-session.aspx

[13] National study report on autonomous shipping – Malaysia, Materials for the Autonomous Shipping in Asia and the Pacific International Conference, 2024, United Nations ESCAP, https://www.unescap.org/sites/default/d8files/event-documents/06Malaysia.pdf

development of regulations and guidelines based on the International Maritime Organization's (IMO) MASS Code.

The training of seafarers is crucial for the successful implementation of autonomous shipping technologies. The Malaysian government has recognized the need for skilled workers to support the development and operation of autonomous ships and has initiated programs to develop a skilled workforce.

Malaysia's strategic location along major shipping routes makes it an ideal candidate for the implementation of autonomous vessels. The Port Klang Authority has initiated several pilot projects to test self-driving vehicles and digital twin technologies[14], focusing on enhancing port operations' safety and efficiency.

In addition to port development, Malaysia is investing in digital infrastructure to support autonomous shipping. The Malaysia Digital Economy Corporation (MDEC) has launched initiatives to promote the adoption of digital technologies in the maritime industry, including autonomous navigation systems.

Policy recommendations for Malaysia

Invest in Technological Infrastructure: To invest in the necessary technological infrastructure to support the development and operation of autonomous ships, including their regulatory frameworks.

Enhance Navigation Safety: To enhance navigation safety by promoting the use of advanced technology such as collision avoidance systems, electronic chart displays, and automated identification systems.

Promote Sustainable Shipping: To promote sustainable shipping practices by encouraging the use of low-emission fuels, such as liquefied natural gas (LNG) and biofuels.

Develop a Skilled Workforce: To develop a skilled workforce to support the development and operation of autonomous ships.

Foster Collaboration: To foster collaboration with other countries, international organizations, & the private sector to promote the safe and sustainable development of autonomous shipping.

[14] Improving Operations: Digital initiatives undertaken by SEA ports, Southeast Asia Infrastructure, May 15, 2024, https://southeastasiainfra.com/improving-operations-digital-initiatives-undertaken-by-sea-ports/

Promising ideas of projects in autonomous shipping in Malaysia

Joint trial project on implementation of autonomous shipping

MISC Berhad is one of the largest Malaysian shipping companies. Cooperation between MISC Berhad and one of the leading technology companies, involved in major projects in major autonomous shipping trial projects, could bring benefits in the spheres of safety and efficiency, environmental sustainability, and the development of advanced marine technologies. Autonomous shipping systems can optimize fuel usage and reduce unnecessary idling, leading to lower greenhouse gas (GHG) emissions. This aligns with MISC Berhad's commitment to reduce the intensity of GHG emissions by 50% in its shipping operations by 203015. MISC has secured a $527 million sustainability-linked loan to finance six very large ethane carriers (VLECs). This loan is structured to align with MISC's decarbonization strategy and includes key performance indicators (KPIs) for environmental performance, incentivizing the company to prioritize sustainability.[16] The joint project on autonomous shipping could help MISC Berhad achieve their zero carbon emission commitments.

[15] Sustainable financing propels MISC's ESG ambitions, 01.09.2022, Lloyds List, https://lloydslist.com/LL1141796/Sustainable-financing-propels-MISCs-ESG-ambitions

[16] MISC inks half a billion dollar sustainability-linked loan, April 20, 2023, by J.O.Mandra, Offshore-energy, https://www.offshore-energy.biz/misc-inks-half-a-billion-dollar-sustainability-linked-loan/

E. THAILAND: NATIONAL EXPERIENCES

Autonomous shipping technologies in Thailand

Thailand is actively exploring the concept of autonomous shipping, driven by the potential benefits of minimizing operational costs, reducing crew difficulties, and lowering the probability of human error-induced incidents during voyages. The country has been involved in research and development prior to commercialization, focusing on the development of autonomous vessel technologies. Thailand has also been influenced by the technological advancements in other countries, particularly in Europe and North America, where autonomous vessel technologies have been studied and regulations prepared for future vessels.

The main organization in Thailand driving the policy initiatives related to robots, autonomous vehicles, and automation systems is the Ministry of Industry, working in cooperation with relevant agencies, including the Office of Industry Economy, Office of Industrial Standard, Department of Industrial Promotion, National Science and Technology Development Agency, and Board of Investment (BOI). In 2017, the Centre of Robotics Excellence (CoRE) was established under the initiative of the Ministry of Industry as a network of leading organizations related to the development of robotics in Thailand. Through collaboration among networking agencies, CoRE aims to enhance the use and the supply of robotics and automation systems by promoting technology transfers from academia to the industrial sector, developing human resources, and developing prototypes of industrial robots, among other initiatives. The Center could play a pivotal role in advancing autonomous shipping and robotics technology for the shipping industry in Thailand[17].

There is not many evidence of the developments in autonomous shipping in Thailand. The most relevant is a small autonomous ship invented in 2015 by the Rajamangala University of Technology Thanyaburi (RMUTT) and Hydro Informatics Institute (HII) with granted financial support from the Ministry of Higher Education, Science, Research and Innovation (MHESI). This is a small remotely controlled autonomous vessel for surveying and collecting data of water quality and maritime resources to minimize human work risks[18]. The autonomous ship can either move to designated locations by using 2.4 GHz radio waves that

[17] Thailand's automation & robotics, site of the Thailand Board of Investment (checked 20.05.2024), https://www.boi.go.th/upload/content/Automation_%26_Robotics.pdf

[18] Waiwit Yodprasit, Autonomous vessels for environmental surveying and data collection, 2015, Ministry of Science and Technology of Thailand, http://oldweb.most.go.th/main/index.php/org/4527.html

pilot the ship remotely or move autonomously by employing a GPS device and inertial measurement unit. The ship, driven by podded propulsion, can reach a maximum speed of 3 meters per second and does not release any noise from the motor while being able to carry up to 90 kilograms of goods of any type.

The legal framework for autonomous shipping in Thailand is still in its early stages. While there are ongoing efforts to develop regulations that support the integration of autonomous technologies, comprehensive national legislation specifically addressing autonomous shipping is not yet fully established. The International Maritime Organization (IMO) is working on developing a Maritime Autonomous Surface Ships (MASS) Code, which is expected to provide a regulatory framework for autonomous shipping globally, including Thailand.[19] Also, the introduction of autonomous ships in Thai water territories will require a series of corrections of the existing Thai Waters Act, comprising the following issues: (a) the duties and responsibilities of the ship master controlling the ship in preventing ship collision and reports to relevant authorities; (b) the mechanism and format of informing the arrival and departure of ships to relevant authorities, including the marine department and port operators; (c) the ability to prevent autonomous ships from entering into the prohibited navigation areas; (d) the withdrawal or confiscation of ship master licenses of autonomous ships that enter into prohibited navigation areas; and (e) the ability to order an autonomous ship to sail only on designated routes and anchor in designated premises.

The training of seafarers in Thailand is still focused on conventional subjects such as cargo handling and stowage, ship communication, IMO conventions, and maritime laws. However, there is a need for improvement in courses and innovative teaching methods to prepare seafarers for the future in fields such as robotics, digitalization, future mobility, and other disruptive innovations. This lack of preparedness in human resources poses a challenge for Thailand in preparing for the adoption of autonomous shipping.

The main barriers and challenges to the introduction of autonomous shipping in Thailand include the lack of preparedness of human resources, the need for improvement in courses and innovative teaching methods, and the regulatory framework, which is still evolving. Additionally, there are concerns regarding cybersecurity, as autonomous vessels are especially vulnerable to cyber threats

[19] Martyn Wingrove, 04 Apr 2023, Autonomous vessels: legislating for a new generation of ships, 2023, https://www.rivieramm.com/news-content-hub/news-content-hub/autonomous-vessels-legislating-for-a-new-generation-of-ships-75634

due to their reliance on complicated computerized systems, large data transfers through satellite links, and satellite navigation.

Policy recommendations for implementing autonomous shipping in Thailand

Recent developments of autonomous shipping in developed countries and policy initiatives ultimately drive Thailand to become a regional leader in automation system industry. Taking into consideration the current constraints in regulatory and human resource aspects, the measures to promote autonomous vessels in Thailand in the medium to long terms are the following:

Establishing forums on autonomous vessels is essential to raise awareness and develop strategies to promote the progressive adoption of autonomous vessels in both domestic and international waters. Specifically, it's crucial to include representatives from the marine transportation sector in the working team to handle the robot, autonomous vehicle, and automation system industry forward. Currently, this industry is primarily led and governed by representatives from the Ministry of Industry. By involving stakeholders from the marine transportation sector, the development gap can be bridged and progress in inventing and commercializing autonomous vehicles for transportation purposes can be expedited.

Maintain an open mindset: the philosophy of launching autonomous vessels into commercial services to save on costs and energy should predominantly focus on the safety management of operations, design, construction, and liability to stakeholders while relevant stakeholders should open their minds to accept new technological advancements.

Innovating education and human resource training and development: the current education courses offered in universities, colleges, and training institutes in the country remain focused on conventional maritime transport subjects but have not yet innovated to cater to changing technological advancements such as robotics, digitalization, future mobility, and other disruptive innovation.

Experiment in a closed system (sandbox). To ensure the commercialization of autonomous vessels, experiments must be conducted in a controlled environment (sandbox). This involves: (a) Designating a Specific Sandbox: Establish a closed water territory for experimentation. Conduct a trial project on low-density routes to ensure safe operations. These routes should connect foreign ports, coastal ports, and Ranong Ports (regional ports operated by the Port Authority of Thailand in the Andaman Sea). (b) Developing Standards: Create standards for autonomous vessels based on internationally accepted guidelines from the

International Maritime Organization (IMO) or global professional engineering societies.

Modernizing regulations: The Thai government should prepare to modernize or introduce regulations related to autonomous vessel design and construction, ship licensing, ship owner and ship controller licensing and qualification, safety management, safety assurance, cybersecurity, ship registration, and ship navigation in water territory based on data collected in sandbox experiments, if applicable.

Technological platform: The government should consider establishing a technological platform to monitor and communicate with autonomous vessels and ship owners/controllers, as well as a mechanism capable of creating reports in emergency or collision circumstances, ship arrival and departures from ports, and liability for ship owners/controllers to relevant stakeholders, taking into account the absence of the master on board the ship.

The national plan for the implementation of autonomous shipping technologies in Thailand should prioritize the development of a comprehensive regulatory framework, the improvement of seafarers' training, and the promotion of public awareness and education on the benefits and challenges of autonomous shipping. The plan should also focus on the development of infrastructure and the integration of autonomous shipping technologies into existing infrastructure and processes. Under the Thai Marine Department, the **draft National Plan of the Implementation of MASS in Thailand** should consist of the following[20]:

1. Raise awareness of policy making, MASS trials research, seafarer & remote operator training (2025 – after non-mandatory MASS Code implementation);
2. Establish working groups (2025);
3. Set up MASS vision (goals, direction): flag state, port state, coastal state, national fleet, maritime institutes (from 2026);
4. Gap analysis to create National Development Strategy Plan: short, medium, long terms;
5. National Development Strategy Plan (NDSP): threat elimination and adequate resource support.

[20] National study report on autonomous shipping – Thailand, Materials for the Autonomous Shipping in Asia and the Pacific International Conference, 2024, United Nations ESCAP, https://www.unescap.org/sites/default/d8files/event-documents/Thailand_national-study-report-on-autonomous-shipping.pdf

Promising ideas of projects in autonomous shipping in Thailand

Promising joint trial project on autonomous shipping

The Thailand Marine Administration has actively participated in and supported workshops and discussions on autonomous shipping implementation in Thailand. Thus, in September 2024, Marine administration held a workshop on autonomous shipping. This included more than 10 ship-owners, including the most valuable in the market: Precious Shipping PCL; Ama Marine PCL; V.L. Enterprise PCL; Siam Lucky Marine Co. Ltd; TMN Co. Ltd.; six training institutions, including the Pattaya Marine Safety Training Center; ship-builders; and ship-management companies. Companies expressed their interest in taking part in autonomous shipping implementations.

Joint training programs for seafarers and joint research with the AMCOL

Collaboration between the AMCOL (Asian Maritime Technological College) and the universities at the forefront of developing training programs for autonomous shipping can bring AMCOL their advanced curricula and training methodologies to enhance the quality of education at AMCOL. This partnership could focus on training operators for autonomous shipping and remote operations centers, which is crucial for the future of maritime transport. Collaborative research and development projects between AMCOL and leading universities will help in the creation of new technologies and solutions for autonomous shipping. These projects will contribute to the advancement of marine technologies and enhance the capabilities of all institutions.

Promising inland water route for autonomous shipping in Thailand

Thailand is part of regional agreements that affect its maritime regulations. For example, the Agreement on Commercial Navigation on the Lancang-Mekong River, signed with China, Laos, and Myanmar, establishes rules for safe navigation and environmental protection. These agreements provide a basis for harmonizing regulations and ensuring the safe operation of autonomous vessels in regional waters. The Eastern Economic Corridor (EEC), which aims to transform the eastern part of Thailand into a hub of technological innovation and logistics, may be a strong framework for the integration of autonomous shipping technologies to improve operational efficiency and reduce environmental impact.

F. VIET NAM: NATIONAL EXPERIENCES

Autonomous shipping technologies in Viet Nam

In Viet Nam, the use of autonomous shipping technologies is still in its early stages. The country has shown interest in automation technology, prioritizing its development across various economic sectors. However, research and development (R&D) projects in the maritime sector are still limited. Current efforts focus on specific ship component systems, such as autopilot systems, rather than comprehensive autonomous shipping technologies. Vietnamese companies such as FPT Software have developed autonomous vehicle software. The experience in automation will be useful for autonomous shipping developments in the future.

Developments in the field of waterborne vehicles in Viet Nam, the research and manufacture of remote-controlled navigating equipment are still limited, with a small size and scale. At the end of 2022, The F.I.R.S.T research teams from the National Economics University and Hanoi University of Technology launched an unmanned vessel to help measure depth, take water samples, and collect garbage in rivers and lakes, with an estimated cost of about US$5,000. The vessel is made of composites, weighs 8 kg, and has dimensions of 1.15 x 0.7 x 0.5 m. The three main functions of the vehicle are depth measurement, water sampling and garbage collection.

The Ho Chi Minh City University of Technology also researched a number of topics, such as design of a dynamic positioning for unmanned surface vehicles using GPS/INS (VIAM-NAVIM) and developed some unmanned equipment models.

The Viet Nam Institute of Geodesy and Cartography has designed and built an unmanned remotely controlled water surface vehicle to conduct surveillance of the seabed and collect the natural resource and environmental data. With the current design, the vehicle has a load capacity of 60kg and is fitted with two batteries, so it can be operated continuously for 8-10 hours.

There are no regulations in governing autonomous shipping in the country. Article 7 of the Viet Nam Maritime Code 2015, (as amended and supplemented in 2018), stipulates that the government encourages research and transfer of advanced and modern technologies in the maritime domain, including developing fleets of seagoing ships, seaports, and the shipbuilding industry through the preferential and incentive policies on taxes and loan interest rates. However, there are no regulations or standards related to autonomous or remotely operated ships as an integrated and independently operated object. The government’s strategy for

sustainable development of Viet Nam's marine economy to 2030, with a vision extending to 2045, emphasizes building a modern and sustainable fleet with high-tech, green, and efficient technologies employing high-quality human resources.

The training of seafarers in autonomous shipping technologies is crucial for the successful implementation of these technologies. However, there is currently a lack of specific training programs or guidelines for seafarers in Viet Nam. Currently, Viet Nam has seven education and training institutions for the maritime field and seafarers: Viet Nam Maritime University, University of Transport of Ho Chi Minh City, Maritime College No.1, Maritime College No.2, Hai Phong Polytechnic College, College of Waterway Transport No.2, and Duyen Hai College. The country has experienced a steady rise in the numbers of seafarers, yet there is a large number of low rank seafarers. An insufficient number of seafarers is still a key challenge to be dealt with. This area requires attention and investment to ensure that seafarers are equipped with the necessary skills to operate and maintain autonomous vessels.

Policy recommendations for implementing autonomous shipping in Viet Nam

Communication and public perception engagement: It is necessary to create and implement a communication campaign or program on autonomous shipping technologies (via public media topics, forums, exhibitions, workshops, press releases), their progress and advantages, and how they could contribute to the fulfilment of the Marine Strategy of Viet Nam by 2030, with the vision extending of 2045. Such activities will draw the attention of all stakeholders, including governmental bodies, economic entities, and research institutes in the maritime field to the new trend of development, hence increasing the support to generate public resources and a consensus perception based on the awareness of the benefits and risks associated with autonomous shipping.

Policy and regulatory frameworks: High-level policies and strategies implemented by the government are supportive to the application of new and modern technologies. However, the maritime regulations need to be revised in a more specific way so that the autonomous shipping technologies can be properly encouraged and developed. The Viet Nam Maritime Code 2015, which is being revised for planned amendment in 2025, should be updated with specific clauses regarding autonomous shipping. The laws guiding regulations such as decrees, circulars, national technical rules for construction and classifications of ships should also be revised accordingly to create a favorable environment in the process of research, adoption, investment and implementation, which cover clear standards for infrastructure, operation areas and routes, communication, safety, security, autonomous vessel design, operation and maintenance, data privacy,

legality and liability, etc. The regulations should be continuously updated based on the gradual maturity of the R&D and experimental results, as well as the updates of international instruments. A roadmap for the application of different levels of autonomous shipping may need to be created in line with the resource and infrastructure conditions available.

Research and Development: There should be a prioritized program with appropriate funding sources (governmental, public, private, or international supporting budgets) for specific research and development projects relating to autonomous shipping technologies, which could be realized in the form of service orders or contracts established with research institutes or capable economic entities. This would be better to encourage partnerships between the maritime industry and economy (such as Viet Nam Maritime Corporation (VIMC), Shipbuilding Industry Corporation (SBIC), Thaco Group, Vin Group) and academia (Viet Nam Maritime University, University Transport of Ho Chi Minh City (HCMC), University of Science and Technology) to develop and test new technologies in a real-world environment, including ones that may be the future beneficiary operators. Collaborations between the domestic entities with potential experience and knowledge in the fields of automation and digital technology, such as Viettel, Phenikaa, Viet Nam Posts and Telecommunications Group (VNPT), and Financing and Promoting Technology Corporation (FPT), are encouraged to deploy new algorithms and sensors, data science, machine learning, and AI application.

Infrastructure: Plans and resource allocation should include considerations for investing in the infrastructure required to support and ensure the efficient, reliable, and secure operation of autonomous shipping technologies. This investment may encompass the construction and maintenance of bases for autonomous ships and equipment, shore-based control centers, autonomous ports and routes equipped with safety navigation facilities, emergency response and rescue systems, high-speed communication networks, and data centers.

Human resource preparation and capacity building: Current training programs in maritime institutions and colleges need to be updated accordingly with formal training on autonomous shipping technologies for the development of a skilled workforce in researching, constructing, operating and maintaining autonomous ships and infrastructure. Viet Nam has a young and skilled pool of IT and data engineers that could timely contribute to the process if additional training can be provided. Collaboration and knowledge-sharing programs from the leading countries in this field is another option.

Collaboration and knowledge-sharing: Collaboration and knowledge-sharing is essential for advancing new technologies such as autonomous shipping. The Government, MoT and Vinamarine should organize a way to connect and share experiences between the domestic stakeholders in autonomous shipping to facilitate the collaboration. This can be a network forum, fan page, portal, or any other appropriate form moderated by Vinamarine. The Government, MoT and Vinamarine should also support in allocating and connecting the international partners with the domestic stakeholders for the same objectives. The international support and cooperation in autonomous shipping technologies, including research and development, funding, co-funding, training, experience sharing, technology transfer based on the win-win principle, benefit harmonizing, and risk sharing should be prioritized.

The implementation of autonomous shipping technologies in Viet Nam (**draft National Plan**) is suggested to be split into 3 phases.

The 1st phase, from 2023 to 2025, is the preparation phase. This phase is focused on carrying out the necessary preparation steps such as developing plans, objectives, projects, awareness and communication, preparing the temporary legal framework, finding resources, connecting partners, setting up environment and infrastructure, etc. for autonomous shipping technologies' research and development.

Phase 2 runs from 2026 to 2030. This phase is for implementing research and development projects on autonomous shipping technologies, as well as for continuing to seek and expand opportunities for collaboration, technology transferring, revising legal and technical regulations based on research results, planning, and infrastructure investment for appropriate levels of automation, including the preparation of port areas and routes for future operation of autonomous ships.

Phase 3 starts from 2030 onward. This phase starts to put into commercial operation pilot autonomous shipping projects at different levels of automation to adjust, plan, and prepare for the replication of this technology in the country's maritime sector.

Promising ideas of projects in autonomous shipping in Viet Nam:

Promising joint trial project on autonomous shipping

Port of Ho Chi Minh City of Viet Nam will be involved in the trials of MASS. The POS Singapore, a 1800 twenty-foot equivalent unit (TEU) containership designed for Level 3 autonomous shipping, is South Korea's first large ocean-going vessel, planned to conduct its first trials on an international shipping route

in September 2024. The vessel incorporates artificial intelligence (AI), Internet of Things (IoT), and "big data" to enable unmanned navigation and digital-based monitoring. The autonomous system includes navigation, communication, security technologies, and advanced digital agency monitoring systems[21]. Vietnamese companies could be involved in Korean companies' trials to get more information about the technology, to cooperate on development and testing of shore-based infrastructure (such as Vessel Traffic Management System (VTMS) or Shore-based Monitoring and Warning (SMW)) in interaction with the MASS, and to prepare Ho Chi Minh City port infrastructure for installation of berthing and unberthing systems.

Joint training programs for seafarers and joint research with the Viet Nam Maritime University

The collaboration between Viet Nam Maritime University (VMU) with other universities at the forefront of developing training programs for autonomous shipping can bring VMU more advanced curricula and training methodologies to enhance the quality of education at VMU. This partnership could focus on training operators for autonomous shipping and remote operations centers, which is crucial for the future of maritime transport. Collaborative research and development projects between VMU and leading universities will help aid the creation of new technologies and solutions for autonomous shipping. These projects will contribute to the advancement of marine technologies and enhance the capabilities of all institutions.

[21] South Korea Launches 1st Container Ship Built for Autonomous Operations, Yonhap News Agency, Marineinsight, 09.05.2024, https://www.marineinsight.com/shipping-news/south-korea-launches-1st-container-ship-built-for-autonomous-operations

III. OTHER ASIA-PACIFIC COUNTRIES' EXPERIENCES

A. THE PHILIPPINES: NATIONAL EXPERIENCE

Autonomous shipping technologies in the Philippines

The Philippines is currently exploring innovative autonomous shipping technologies, which include better control of the ship's position and speed, efficient communication with external systems, integrated control of on-board navigation systems, ship management through remote support and control, and risk management using advanced technologies. These technologies are being developed and tested by various stakeholders, including shipping companies, shipowner's associations, investors, universities, and R&D companies[22].

The Philippines has no current law on autonomous shipping, but the incumbent President has shown enthusiasm in strengthening the Philippine maritime industry. The country is also part of several international maritime conventions, which provide a framework for the development and implementation of autonomous shipping technologies.[23]

Implementation of innovative autonomous navigation technology is one of the priorities for maritime industry in the Philippines. Maritime Industry Development Plan of the Republic of the Philippines for 2019 – 2028 covers eight priority programs that focus on the enhancement of maritime transport security; promotion of environmentally sustainable maritime industry; implementation of maritime innovation and digitalization, among others. The Plan features ten (10) priority programs to be implemented till 2028. In particular:

Program 8: Maritime Innovation and Knowledge Center with a focus on improving capacity of MARINA and other stakeholders to innovate and operate modern technologies and implement best practices in shipping.

Program 9: Development Of Competitive And Highly Skilled Filipino Maritime Professionals: Development of maritime programs for the commercial operation of Maritime Autonomous Surface Ships (MASS); and Upgrading the competence of domestic seafarers. Major program components:

- Component 4: Development of maritime programs for the commercial operation of Maritime Autonomous Surface Ships (MASS) and for other

[22] Dennis A. Pandeagua, The significance of marine autonomous surface ships: prospects to the Philippine maritime industry, dissertation, 2019, World Maritime University, https://commons.wmu.se/cgi/viewcontent.cgi?article=2227&context=all_dissertations

[23] Adopting the ten-year maritime industry development plan 2028, Executive order no.55, by the President of the Philippines, January 02, 2024 https://marina.gov.ph/maritime-related-laws/

career options.

- 4.1 Studies and Research on the Operation of MASS.

Lead in undertaking research and operations of Maritime Autonomous Surface Ships including the development of training and testing of training programs.

Legislation on autonomous shipping in the Philippines

The current legal framework for autonomous shipping in the Philippines is still evolving. The Philippines has not yet developed specific laws or regulations addressing autonomous shipping technologies. However, the country is actively participating in international efforts to develop a regulatory framework for autonomous shipping through the International Maritime Organization (IMO) and other regional organizations. The IMO has been actively working on developing a regulatory framework for Maritime Autonomous Surface Ships (MASS) through the Maritime Safety Committee (MSC) and the Legal Committee (LEG). The IMO has defined MASS as a ship that can operate to some degree without a human onboard, and it has identified four degrees of automation, ranging from automated processes and decision support to fully autonomous ships. The Philippines, as a signatory to several international maritime conventions, is expected to implement the IMO's regulations and guidelines on autonomous shipping. The country is also participating in regional initiatives, such as the "International Conference on Autonomous Shipping in Asia and the Pacific", which aims to raise awareness of relevant stakeholders about benefits and developments in autonomous shipping technologies and develop policy recommendations for the introduction of autonomous navigation in the region. In summary, while there is no specific legal framework for autonomous shipping in the Philippines, the country is actively participating in international efforts to develop a regulatory framework for autonomous shipping and is expected to implement the IMO's regulations and guidelines once they are finalized.

The training of seafarers is crucial for the successful implementation of autonomous shipping technologies. The Philippines should strengthen its maritime curriculum to address the needs of the future work in maritime trends, focusing on STEM education and preparing the infrastructure of cyber connectivity and autonomous support systems[24].

The Department of Transportation (DOTr) is the lead agency for the implementation of e-Navigation in the Philippines, with specific components of

[24] Dennis A. Pandeagua, The significance of marine autonomous surface ships: prospects to the Philippine maritime industry, dissertation, 2019, World Maritime University, https://commons.wmu.se/cgi/viewcontent.cgi?article=2227&context=all_dissertations

e-Navigation assigned to the Philippine Ports Authority (PPA), the Philippine Coast Guard (PCG), and the Maritime Industry Authority (MARINA). The DOTr is assisted by the attached agency, and each agency co-leads on e-Navigation related functions which the agency has their respective operational expertise in relation to autonomous shipping components.[25]

Proposals for Implementation of Autonomous Shipping National Plan

The proposals for implementing a national plan for autonomous shipping technologies in the Philippines emphasize a comprehensive and unified approach. This includes leveraging existing government agencies and developing a strategic plan based on regional experiences. According to a survey conducted on February 28-29, 2024, the Philippines' representatives have identified the following key steps for the implementation of autonomous shipping:

Establish Comprehensive Safety Standards. Developing robust safety standards and regulations is essential to ensure the secure operation of Maritime Autonomous Surface Ships (MASS). Balancing innovation with safety is crucial for the acceptance and integration of autonomous vessels into maritime operations.

Training and Skill Development. Prioritizing the development of specialized training programs for seafarers and maritime professionals is vital. This ensures that individuals responsible for operating and managing MASS possess the necessary skills and competence to navigate the complexities of autonomous technology.

Robust Cybersecurity Measures. Implementing strong cybersecurity protocols is a top priority. Since MASS relies heavily on digital systems and connectivity, safeguarding against cyber threats is critical to prevent unauthorized access, potential system breaches, and to ensure the integrity of autonomous operations.

Promote Environmental Sustainability. Integrating MASS into a framework that promotes environmental sustainability is imperative. Prioritizing technologies that enhance fuel efficiency, reduce emissions, and uphold environmental responsibility aligns with global efforts to make shipping more eco-friendly.

Foster International Collaboration. Setting global standards and fostering international collaboration for MASS is crucial. A harmonized approach ensures

[25] Orly Wong, A study on the e-navigation government framework: a Philippine perspective, 2023, World Maritime University Dissertations, https://commons.wmu.se/cgi/viewcontent.cgi?article=3294&context=all_dissertations

consistency, facilitates cross-border operations, and enhances the acceptance of autonomous vessels on a worldwide scale.

Enhance Public Awareness. Addressing public perception and gaining societal acceptance of MASS is a priority. Public awareness campaigns, transparent communication, and demonstrating the safety and benefits of autonomous shipping are essential to building trust and overcoming potential resistance.

These priorities highlight key areas that need attention to ensure the successful integration of Maritime Autonomous Surface Ships into the maritime industry. The first priority for the Philippines' policy on implementing autonomous navigation, according to experts' opinions will focus on coastal navigation.

IV. SET OF POLICY RECOMMENDATIONS REGARDING AUTONOMOUS SHIPPING IN ASIA AND THE PACIFIC

A. POLICY RECOMMENDATIONS FOR NATIONAL PLAN DEVELOPMENT

Taking into account the reports of international and regional experts, as well as the results of workshops and the conference, it can be concluded that a number of general recommendations are applicable for the development of national plans on the introduction of autonomous shipping in the target countries.

1. Establishing of policy and regulatory frameworks. The IMO will provide the International MASS Code as a common framework for the implementation of autonomous shipping. Thus, governments of the countries need to adopt the new Code through introduction of related nation regulations. At the same time, to accelerate the introduction of new technologies and eliminate the technological gap from developed countries, the governments of the target countries need to develop clear and supportive policy and regulatory frameworks that encourage the development and adoption of autonomous shipping technologies. This includes creating a favorable investment climate, providing tax incentives and subsidies, and establishing clear rules and standards for safety, security, and data privacy, as well as the development of insurance policies covering risks associated with autonomous shipping, including damage to vessels, cargo loss, and liabilities.

2. Training and skilling development. Developing a skilled workforce is essential to implementing autonomous shipping technologies, especially considering the role of the target countries in the global supply of seafarers. The government and universities of the target countries have to provide training and skilling programs for seafarers to adapt to new technologies and roles in autonomous shipping. The target countries also need to update education courses to incorporate changing technological advancements and disruptive innovations.

3. Universities' research and development. The significant role of universities and other academic institutions in the current development of autonomous shipping is to be funded and supported by relevant R&D support programs, which, together with training programs, will create the necessary technological foundation in the target countries. This should also include fostering of partnerships between government agencies, industry stakeholders, and research institutions to leverage expertise and resources.

4. Public engagement and awareness. A national plan on implementation of autonomous shipping should include efforts to engage the public and build both

industry and society support for these technologies. This may include establishment of regular national and international forums, public consultations, and communication campaigns to engage the public and stakeholders through regular updates and educational initiatives to ensure understanding and address challenges. As a result, significant value could be contributed from engagement with international organizations by sharing best practices, collaborating on research and development, and ensuring alignment with global standards.

5. Investment in infrastructure and sandboxes. The development of the entire autonomous shipping eco-system would require risky upfront investments, which are difficult to shift only to the private sector. To maintain advanced technology in the target countries, every government needs to invest in the development of infrastructure supporting autonomous shipping, including communication, cybersecurity, e-navigation and smart ports, fostering public-private partnerships to build capacity within the maritime sector to design, develop, and operate autonomous vessels. This should include investment in sandboxes (testbeds and trial projects), enabling the sector to test these technologies in a real-world environment and for national stakeholders to gain the required experience, including the private sector.

6. Establish monitoring and evaluation frameworks to track progress and impact, identifying areas needing improvement.

These recommendations provide a comprehensive framework for the successful development of national plans on implementation of autonomous shipping technologies, ensuring safety, sustainability, and efficiency in maritime operations.

B. POLICY RECOMMENDATIONS FOR REGIONAL COLLABORATION

Along with national programs for the introduction of autonomous shipping, the similarity of the current state and goals of the target countries also makes it possible to develop recommendations for international regional cooperation among ESCAP member states. These recommendations will also contribute to the implementation of national plans on introduction of autonomous shipping in the target countries.

The key aim is to foster collaboration and partnerships. This involves actively engaging regional stakeholders to promote the safe and sustainable development of autonomous shipping by the following means:

1. Establishing forums or platforms for knowledge exchange

International and Regional Stakeholder Forums: Establishing forums or platforms where international and regional stakeholders can connect and exchange knowledge. This can facilitate ongoing collaboration and ensure alignment with global standards and objectives. By working together, countries can use each other's strengths and experiences to create a supportive environment for the development and implementation of autonomous shipping technologies on an international and regional scale. Examples include the International Ship Autonomy and Sustainability Summits[26] and the Maritime Autonomous Systems Regulatory Working Group Conference[27].

Workshops and Conferences: Organizing workshops and conferences can raise awareness and build capacity among stakeholders. These events can provide a platform for sharing best practices and discussing the latest developments in autonomous shipping. The ESCAP workshops on autonomous shipping are an example of such initiatives.

2. Joint research and training projects

Collaborative Research Initiatives: Engaging in joint research and development initiatives to share expertise, resources, and best practices. This collaboration can address common challenges and accelerate progress in the adoption of autonomous shipping technologies. For instance, the collaboration between SINTEF Ocean and TCOMS in Norway and Singapore focuses on advancing

[26] The 5th International Ship Autonomy and Sustainability Summit, 5 June, 5, 2023, https://nor-shipping.com/autonomy-summit-2023/ (Checked 20.05.2024)

[27] MASRWG Conference 2024, January,16-17, https://www.maritimeuk.org/priorities/innovation/maritime-uk-autonomous-systems-regulatory-working-group/masrwg-conference-2024/ (Checked 20.05.2024)

smart and autonomous maritime systems through joint research projects.[28] Additionally, the establishment of the Joint Laboratory on Smart and Autonomous Navigation Technologies for Ships by Chinese, Korean, and Russian universities serves as a model for international cooperation in this field.[29]

Universities cooperation: Foster universities cooperation in training and joint R&D projects, including dedicated scientific programs funded by target countries. Universities can play a significant role in developing new technologies and training the future workforce.

Training Programs: Developing training programs to build the capacity of maritime professionals to operate and maintain autonomous vessels. This includes updating maritime curricula to incorporate new technologies and providing hands-on training through simulators and real-world trials. The MASSPeople initiative, which focuses on developing training standards for the maritime workforce, is a relevant example.[30]

3. Participation in the development of legal frameworks for MASS

Participation in the development of legal frameworks for MASS at the platform of the International Maritime Organization (IMO) to regulate MASS and to ensure these autonomous ships operate safely and in coexistence with conventional ships.

During the 108th session of the IMO's Maritime Safety Committee (MSC) from May 15 to 24, 2024, significant progress was made in developing the draft MASS Code. The Committee noted the progress in restructuring chapters and refining draft provisions. However, it was acknowledged that additional work is required to finalize the Code. Consequently, the Committee revised the Road Map for the development of the MASS Code as follows[31]:

May 2025: Finalize and adopt the non-mandatory MASS Code to bring it to the experience-building phase by the first half of 2026.

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[28] Autonomous shipping and human training: Where we stand, Safety4Sea, September 26, 2022, https://safety4sea.com/cm-autonomous-shipping-and-human-training-where-we-stand/ (Checked 20.05.2024)

[29] Joint Lab on smart and autonomous ships created by Chinese, Korean, and Russian universities, Cyprus shipping news, March 11, 2022, https://cyprusshippingnews.com/2022/03/11/join-lab-on-smart-and-autonomous-ships-created-by-chinese-korean-and-russian-universities/ (Checked 20.05.2024)

[30] MASSPeople initiative, https://www.masspeople.org/

[31] Maritime Safety Committee - 108th session (MSC 108), 5-24 May 2024, https://www.imo.org/en/MediaCentre/MeetingSummaries/Pages/MSC-108th-session.aspx

2028: Commence the development of the mandatory MASS Code, based on the non-mandatory Code, and consider amendments to SOLAS (new chapter) for the Code's adoption.

By 1 July 2030: Adoption of the mandatory Code, for entry into force on 1 January 2032.

Active target countries collaboration within the framework of the IMO's committees and working groups are very important. The most valuable is participating in the Joint Working Group and the intersessional MASS Correspondence Group, closely working on the development of the MASS Code. By working together, countries can harmonize their regulatory approaches and contribute to the development of a robust and comprehensive MASS Code. Also, participation at the IMO's committees and Working Groups on MASS are valuable for driving the implementation of new autonomous navigation technologies.

ESCAP can serve as one of the international platforms, mentioned above, for the forums and projects.

V. POTENTIAL JOINT PROJECTS AND KEY TECHNICAL SOLUTIONS FOR THEIR IMPLEMENTATION

Autonomous shipping is rapidly expanding, particularly in the Asia-Pacific region, but in most of the cases, the comprehensive projects are implemented inside of the countries without or low external cooperation. These projects focus on various aspects of autonomous shipping, including regulatory frameworks, trial projects, and technological development. Below are examples of the most significant ongoing national projects.

The Korea Autonomous Surface Ship Project (KASS) project involves the development of autonomous navigation systems, including collision avoidance and decision support systems. It integrates navigation data to enhance the safety and efficiency of autonomous vessels. The project also focuses on creating a regulatory framework that addresses legal and liability issues, ensuring the safe operation of autonomous ships. International cooperation is limited to communications, such as a technology exchange meeting between the Korea Research Institute of Ships & Ocean Engineering (KRISO) and Lloyd's Register (LR) in London, UK, to discuss technological development and verification of autonomous ships.

The MEGURI2040 project in Japan involves the development of autonomous navigation systems, including collision avoidance and decision support systems. The project also utilizes advanced sensors and sensor fusion techniques, such as video analytics, radar processing, and high-precision positioning systems. Connectivity solutions, including communication systems and cybersecurity measures, are integral to the project's success. The project aims to commercialize fully autonomous ship technology by 2025, with trial voyages and the development of ship-shore operations. The project primarily involves Japanese companies and organizations.

The ARNTP, also known as the a-Navigation Trial Project, is a significant initiative in Russia involving the trial of maritime autonomous surface ships (MASS) in commercial operations. The project includes the development and testing of autonomous navigation systems, such as the Sensor Fusion Module (SFM) and the Automatic Collision Avoidance Module (ACAM). These systems integrate navigational data from various sources, including radar, AIS, and optical systems, to ensure safe and efficient navigation. The project also focuses on connectivity solutions, including secure communication systems and remote control capabilities. The ARNTP has led to the development of national regulations and the approval of technical solutions for autonomous navigation by

the Russian Maritime Register of Shipping. There is no specific mention of international cooperation in the ARNTP project.

This examples of the projects highlighted disproportion in the development and even access to the new technology between different countries of the region. We see one group of developed countries leading in autonomous navigation implementation in the Asia Pacific region (China, Korea, Japan, Singapore, Russia) and another group without access to this technology (mentioned above target countries). This creates both a growing imbalance in the regional maritime industry and risks of losing competitiveness of these countries in the global maritime industry.

Taking into account the best practices, as well as the readiness target countries to cooperation it is possible to determine the promising areas of joint projects, as well as key technical solutions.

The key areas for joint projects are:

Exchange of experience and support for the implementation of regulatory frameworks: As an example, this may be a permanent forum (including within the framework of the ESCAP research project) for the exchange of experience, harmonization of approaches, and joint development of solutions to problems of implementing international regulation of MASS, taking into account regional and country specifics. Such a forum would join experts from maritime administrations, universities and other institutions involved in the development of national MASS regulations.

Joint training programs for seafarers on autonomous shipping: Joint development of training programs, exchanges of teachers and students, introduction of advanced learning technologies, including new simulators for new specialties (remote control operators, MASS crews, Vessel Traffic Services (VTS) operators, etc.). Potential participants of these programs could be universities of the target countries like Indian Maritime University (IMU), Asian Maritime Technological College (AMCOL), and Viet Nam Maritime University (VMU), as well as universities of countries leading in the implementation of autonomous shipping (Korea, Russia, or China).

Support for the implementation of trial projects (sandboxes) and the development of technological solutions: Development of local knowledge and technologies is vital both for national competitiveness and the maturity of the new technology in the international market. Joint projects in this area should encourage knowledge sharing and technology transfer between leading countries and target countries, involving all industry stakeholders: shipping companies (like

MISC Berhad or PT Pelayaran Nasional Indonesia), technology companies (like Institute Teknologi Sepuluh Nopember or Cochin Shipyard), ports and inland waterways (like Port of Ho Chi Minh City or administration of Lancang-Mekong River), research institutions, and universities. Governments play a critical role with regulating and funding such collaborative projects aimed at transfer and development of their own technical solutions.

The following key technical solutions should be in the focus of joint projects:

- Autonomous Navigation Systems, including collision avoidance software, decision support software and integration of navigation data;
- Advanced sensors and sensor fusion techniques[32], including video analytics and radar processing based on machine learning technologies, affordable IR cameras, lidars, high-precision positioning systems, and other sensing components;
- Connectivity solutions, including communication systems, cybersecurity, remote data services. and automatic exchange of V2V data;
- Solutions for new segments of autonomous shipping like autonomous ports or inland fleets, including shore-based infrastructure, auto-berthing systems, real-time hydrographic and weather data, and adopted vessel traffic management systems;
- New simulators for training of personnel operating MASS, including remote and VR/AR technologies.

Supporting projects and developments in these areas, which are new to the world market, can ensure the development of their own technology companies and increase the competitiveness of the maritime industry as a whole for all Asia-Pacific countries.

[32] Sensor fusion techniques - is the process of integrating data from multiple sensors to produce more accurate, reliable, and comprehensive information than what could be obtained from individual sensors alone. This process reduces uncertainty and enhances the functionality of systems in various applications, such as robotics, autonomous vessels, and industrial automation.

List of abbreviations

ACAM – Automatic Collision Avoidance Module

AI – Artificial Intelligence

AIS – Automatic Identification System

AMCOL - Asian Maritime Technological College

a-Navigation – Autonomous Navigation

ANS – Autonomous Navigation System

APEC - Asia-Pacific Economy Cooperation forum

APSN - Asia-Pacific Economy Cooperation forum Port

AR/VR - Augmented Reality/Virtual Reality

ARNTP – Autonomous and Remote Navigation Trial Project

BMIC - Batam Maritime Industrial Center

BOI - Board of Investment

CEMS - Center for Excellence in Maritime and Shipbuilding

CCTV – Closed Circuit Television

CGT – compensated gross tons

CoRE - Centre of Robotics Excellence

COLREG – Convention on the International Regulations for Preventing Collisions at Sea

DGST - Directorate General of Sea Transport

DOTr - Department of Transportation

DSIR - Department of Scientific and Industrial Research

DWT – Deadweight Tons

EDI – Electronic Data Interchange

EEC - Eastern Economic Corridor

FPT – Viet Nam Financing and Promoting Technology Corporation

GHG - Greenhouse Gas

HCMC – Viet Nam Ho Chi Minh City

HII - Hydro Informatics Institute

I-Boat - Intelligent Boat

LiDAR - Light Detection and Ranging

LNG - Liquefied Natural Gas

IMO – International Maritime Organization

IoT - Internet of Things

IRCLASS - Indian Register of Shipping

ITS - Institut Teknologi Sepuluh Nopember

KASS – Korea Autonomous Surface Ship Project

KRISO – Korea Research Institute of Ships and Ocean Engineering

MASS – Maritime Autonomous Surface Ships

MDEC - Malaysia Digital Economy Corporation

MISC Berhad – Malaysian International Shipping Corporation

MMEA - Malaysian Maritime Enforcement Agency

NCoEGPS - National Centre of Excellence for Green Port & Shipping

NMICPS - National Mission on Interdisciplinary Cyber-Physical Systems

OSA – Optical Surveillance and Analysis System

PELNI - PT Pelayaran Nasional Indonesia

RMI - Robo Marine Indonesia

RMUTT - Rajamangala University of Technology Thanyaburi

POS - Port of Singapore

PT PAL - PT PAL Indonesia (Persero)

R&D – Research and Design

RCC – Remote Control Centre

RCS – Remote Control Station

RS – Russian Maritime Register of Shipping

SBIC – Viet Nam Shipbuilding Industry Corporation

SFM – Sensor Fusion Module

SMW - Shore-based Monitoring and Warning

SOLAS – International Convention for the Safety of Life at Sea

TEU – Twenty-foot Equivalent UNIT

TiHAN - Technology Innovation Hub on Autonomous Navigation

TSS - Traffic Separation Scheme

VIMC – Viet Nam Maritime Corporation

VMU - Vietnam Maritime University

VNPT – Viet Nam Posts and Telecommunications Group

VTS - Vessel Traffic Services

VTMS – Vessel Traffic Management System

USV – Unmanned Surface Vehicle